Sweden: energy security, economic opportunity and partnership priorities

POLICY MEMO · COUNTRY INSIGHTS · ALLIED INDUSTRIAL BASE EXPLORER · NEIS CENTER

AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- Sweden is most exposed in downstream gas, upstream coal and upstream oil, where its Energy Security is higher than only 7%, 14% and 18% of the other economies scored. Gas use per person is low and falling, and coal and oil production fall short of consumption.[1]
- Its strongest openings are in midstream gas, midstream nuclear and downstream wind, where its Economic Opportunity is higher than 92%, 92% and 91% of the other economies scored, helped by exports, investment and output.[1]
- On the paper's weighting, the Strategic Index puts upstream batteries 1st among Sweden's own sectors, 0.015 ahead of the next; policy activity places it in the group with the most policy.[1]
- The Partnership Strength Index rates Finland highest for Sweden, as a reliable supplier; it is also Sweden's 3rd trade partner, while China, its 4th, ranks 44th on Partnership Strength.[1,2]

1. Energy security risks and why

8 of Sweden's 30 placed sectors, those where both scores can be compared with other economies, sit in the two low-energy-security corners. Compared with other economies in each sector, its Energy Security is lowest in downstream gas, upstream coal, upstream oil and downstream oil, all in the Secure corner. It is higher than only 7%, 14%, 18% and 29% of the other 98, 95, 96 and 98 economies scored, with scores of 0.35, 0.26, 0.37 and 0.51. The paper's framework associates that corner, which pairs low opportunity with low energy security, with securing supply through partners.[1]

In downstream gas, energy access pulls the score down: over the years to 2025, gas consumption per person changed by -16.7%, to 2.87 GJ per person. In upstream coal and upstream oil, import needs pull the score down: production minus consumption was -0.0608 EJ for coal and -0.52 EJ for oil in 2025. Both figures take production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately. In upstream oil, reserves also pull it down: no oil reserves are recorded, which the index counts as zero.[3,4]

The data mark downstream solar, midstream nuclear and midstream wind as its high-priority sectors, all in the Upgrade corner. Their Energy Security is higher than 37%, 47% and 48% of the other economies scored. In downstream solar, reserves pull the score down: on solar resource potential, Sweden scores higher than only 12% of the other economies. In midstream nuclear, trade pulls it down: on its trade deficit in these goods, 0.1% of GDP in 2023, it scores higher than only 3% of the other economies.[1,5]

For context, its largest suppliers in upstream oil in 2024 were Norway (35.3% of imports, which totalled USD 17 bn) and the United States (9.7%). In upstream coal, Australia supplied 59.1% of its imports.[2]

2. Economic opportunities and why

24 of the 30 placed sectors sit in the two high-opportunity corners, where Economic Opportunity is above the sector's middle economy. Compared with other economies in each sector, its Economic Opportunity is highest in midstream gas, midstream nuclear and downstream wind. It is higher than 92%, 92% and 91% of the other 155, 153 and 92 economies scored, with scores of 0.62, 0.58 and 0.55. Of these, midstream nuclear sits in the Upgrade corner and the others in the Grow corner.[1]

Trade pushes the score up most in midstream gas and midstream nuclear. Sweden reported exports of USD 2.32 bn and USD 827 m in 2024, with revealed comparative advantages of 1.66 and 4.74. In midstream nuclear, Sweden held 2.9% of world exports in 2023. For context, France and Ukraine took 39.2% and 36.9% of its exports in this sector in 2024 (reconciled trade data).[2,5,6]

In downstream wind, investment and production push the score up. Production was 38.7 TWh in 2025, and on annual investment Sweden scores higher than 87% of the other economies.[1,3]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by combining opportunity, security need, policy activity, the supply-chain stage and the technology's low-emission value. On the paper's weighting, it puts upstream batteries 1st, downstream batteries 2nd, upstream nuclear 3rd, upstream electric grid 4th and upstream wind 5th among Sweden's own sectors. The leader, upstream batteries, scores 0.67, 0.015 ahead of the next. The sectors that follow are almost tied, with gaps of 0.003, 0.008, 0.006 and 0.000. In downstream batteries, the policy term adds 0.200; without it, the sector falls into the middle third.[1]

The allied pick, the leading upstream or midstream sector after a screen for a resource base, is upstream nuclear (0.65), 0.014 ahead of upstream wind. The screen leaves out upstream batteries: no resource base is recorded. The pick sits in the Watch corner (low opportunity, high energy security).[1,7]

Comparing priority, the index without its policy term, with policy activity, upstream batteries, upstream nuclear, upstream electric grid, midstream batteries and upstream wind combine top priority with the most policy. No top-priority sector falls in the group with the least policy, while downstream wind combines bottom priority with the most policy. The sectors with bottom priority and the least policy are upstream oil, upstream coal, midstream gas, midstream oil, midstream coal, downstream oil and downstream coal. Sweden carries the Climate Action Tracker rating Insufficient (November 2024) and has 221 records in the International Energy Agency's Policies and Measures database, 103 of which name a technology.[1,8,9]

4. Partnership priorities and why

The Partnership Strength Index rates partners as suppliers (Friendshore), as markets for exports (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. From Sweden's viewpoint it ranks Finland 1st, Norway 2nd, the United States 3rd, Germany 4th and Denmark 5th of 191 partners. Finland, the United States and Germany stand out most as a reliable supplier. Germany's best sectors as a supplier, downstream solar, downstream batteries, upstream green hydrogen, midstream nuclear and midstream wind, include each of Sweden's high-priority sectors. Norway and Denmark stand out most as an export market; Norway's best sectors as a market are upstream electric vehicles, upstream batteries, midstream batteries, midstream solar and midstream wind.[1]

These priorities match today's ties. Germany is Sweden's 1st trade partner on average across its energy-goods baskets, Norway its 2nd, Finland its 3rd, the United States its 5th and Denmark its 10th. Each is a priority partner with strong ties today and votes in the same bloc at the UN. Finland, Norway, the United States and Germany also share membership of NATO and the Minerals Security Partnership with Sweden. Poland, its 6th trade partner, ranks 13th on Partnership Strength. By contrast, China is its 4th trade partner but ranks 44th, and they vote in opposite blocs at the UN.[1,2,10,11]

As a partner to the coalition, Sweden ranks 12th of 192 economies, on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads production of each upstream and midstream sector across coalition members, its largest designed share is 1.5% of midstream nuclear.[1,7]

5. Data notes and caveats

Country Insights shows these values as of 12 October 2026. Scores compare Sweden with other economies in the same sector; they are not measures of risk or profit in absolute terms. All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings. Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order. Some figures behind these scores are filled in where the source has no value: with zero where a source lists every holder, otherwise with the world average. Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up. In the allied network design the European Union is one producer; Sweden's shares are derived from its weight within the EU. The Climate Action Tracker rates the EU as a whole; Sweden carries the EU's rating. Sweden's counts leave out the EU's own 438 International Energy Agency policy records. Trade figures count trade with other EU members.

Sources

1. Energy Security, Economic Opportunity, Strategic Index and Partnership Strength Index, paper baseline (Opportunity Security Indices pipeline); CC BY 4.0 (the paper's indices).
2. United Nations, UN Comtrade Database, HS6 bilateral trade (all reporters), 2020 and 2024; reconciled by the authors (pulled 2026-10-07 to 2026-10-09); attribution; https://comtradeplus.un.org/.
3. Energy Institute, Statistical Review of World Energy 2026 (data to 2025); attribution; https://www.energyinst.org/statistical-review.
4. U.S. Energy Information Administration, International Energy Statistics, bulk file INTL.zip: crude oil incl. lease condensate, NGPL, dry natural gas and coal production; petroleum and other liquids, dry natural gas and coal consumption, bulk file dated 2026-10-03 (oil production to 2025; gas and coal to 2024); open; https://www.eia.gov/international/data/world.
5. Growth Lab at Harvard University, The Atlas of Economic Complexity, HS92 6-digit country-product-year data (data to 2023); attribution; https://atlas.hks.harvard.edu/.
6. United Nations, UN Comtrade Database, exports to World by reporter, 2024 (2024 data); attribution; https://comtradeplus.un.org/.
7. Allied network design (the paper's model design of coalition production shares).
8. Climate Action Tracker, country ratings (ratings updated 2022-2025); attribution; https://climateactiontracker.org/.
9. International Energy Agency, Policies and Measures Database (export of 2026-09-30); attribution; https://www.iea.org/policies.
10. Bailey, M., Strezhnev, A. and Voeten, E., United Nations General Assembly Voting Data, ideal points (Harvard Dataverse V39.0) (sessions to 2023); attribution; https://dataverse.harvard.edu/dataset.xhtml?persistentId=hdl:1902.1/12379.
11. Economic-security agreements: G7, NATO, Indo-Pacific Economic Framework, Minerals Security Partnership (and Forum), U.S. free-trade agreements (compiled by the authors from official membership lists).
